

Leveraged Wrong-Way Risk

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Abstract

We introduce a simple model for the credit exposure to leveraged and collateralized counterparties. Wrong-way risk is captured by linking the counterparty default probability directly to changes in the portfolio value. This applies e.g. to leveraged firms such as hedge funds where large collateral calls can be the driver of default. We show that our model is able to reproduce the large losses observed in recent events. These losses were unexpected based on typical exposure models which neglect the relatedness of large market moves and the viability of the counterparty. Our model is intuitive to parameterize and straightforward to implement and thus provides a useful tool for assessing the credit risk inherent in leveraged portfolios.

1 Introduction

Recent events have shown that standard models for counterparty credit exposure are not well suited to capture the risk inherent in highly leveraged and collateralized portfolios. High levels of collateral, in particular in the presence of initial margin (IM), mean that modeled exposure is negligible. However, if leverage is significant then the counterparty will be sensitive to large market moves. In particular, it is often the associated large collateral calls that are the proximate cause of default. This relatedness between the counterparty default and large changes in its portfolio value can lead to large and unexpected losses.

This is typical e.g. for hedge funds, with the most recent and significant example being the well-documented default of Archegos Capital Management [1] which resulted in losses totaling ca. \$10 billion across its counterparties. Other similar cases are discussed in [2] and references therein. The defining features are the direct link between extreme market moves and subsequent large collateral calls leading to the counterparty defaults and losses far in excess of any model predictions.

In order to capture the risk posed by leveraged counterparties we need to model the relatedness between the default event and the portfolio dynamics. This is the well-known wrong-way risk (WWR) effect. WWR is often neglected in exposure models and in cases where it is

considered, it is typically modeled by correlating the counterparty credit spreads and the market risk factors underlying the counterparty portfolio.

In this article we explore a different mechanism and we model the default probability directly as a function of the change in portfolio value over the margin period of risk. This relatedness does not require any intermediate transmission via market factors. We thus directly incorporate the feature that large collateral calls due to outsized moves in the portfolio value will lead to counterparty default.

The literature regarding this specific type of wrong-way risk is limited. The earliest analysis can be found in Hull and White [4] where the counterparty default intensity is modeled as an exponential function of the portfolio value. Two more recent articles are by Dickinson [2] and Anfuso [3]. Dickinson models the risk of large sudden moves in the portfolio value via Levi jump processes which are correlated to the counterparty default probability. Anfuso on the other hand takes a bottom-up approach to model the balance sheet of a leveraged counterparty in order to directly link the solvency of the firm with its portfolio dynamics. In both approaches the authors show that WWR has a large impact on the modeled exposures.

The challenge with these types of WWR models lies in the calibration which can be difficult. In this article we explore what is perhaps the simplest example of such a model which also has an intuitive parameterization. This is to gain intuition around the model behaviour and to assess whether it is possible to provide estimates of exposure that are commensurate with the losses observed in recent events.

Our model can be viewed as a special case of the Hull-White approach, where we define the default intensity as a function of the *change* in portfolio value over the margin period of risk. The counterparty default occurs when the change in value exceeds a counterparty specific threshold. The distribution of portfolio returns can in principle be chosen arbitrarily, however we show that using the common student's t-distribution leads to realistic results and allows for analytic expressions for the exposure quantities of interest.

2 Exposures Conditional On Default

The counterparty exposure at a future time t is defined as the potential loss or claim in the event of the counterparty default. By taking the expectation of the exposure we obtain the expected exposure profile $EE(t)$ which is a fundamental quantity for XVA and counterparty capital calculations. A key point for the consideration of wrong-way risk is that the expected exposure is defined as an expectation *conditional* on default. Here we recall some basic definitions and background that we will make use of in later sections. For further details we refer to [5].

In practice, default events are not instantaneous but will occur over a certain grace period which is often referred to as the margin period of risk (MPoR) and which we will denote by Δ . If the counterparty is subject to a collateral agreement (CSA) then Δ represents the time between the last collateral posting and the unwind of the defaulted portfolio.

We can hence write the expected exposure at a future time t as:

$$EE(t) = \mathbb{E}_0 \left[(V_{t+\Delta} - C_t)^+ | t < \tau < t + \Delta \right] \quad (1)$$

where V_t is the portfolio value, C_t is the collateral and τ is the default time of the counterparty. The expectation $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot | \mathcal{F}_t]$ is defined with respect to the filtration $\mathcal{F}_t$ and we denote today by $t = 0$. Here and in the following we set interest rates to 0 for simplicity.

It is also well-known that this expectation can be expressed in terms of “path-weights” ω_t as follows:

$$\mathbb{E}\mathbb{E}(t) = \mathbb{E}_0 [(V_{t+\Delta} - C_t)^+ \omega_t] \quad (2)$$

where

$$\omega_t = \frac{P(t < \tau < t + \Delta)}{\mathbb{E}_0 [P(t < \tau < t + \Delta)]} \quad (3)$$

is just the (scenario dependent) default probability normalized by the expected default probability. It is evident that in the case that the default probability and the portfolio value are independent the exposure reduces to the standard expression:

$$\mathbb{E}\mathbb{E}(t) = \mathbb{E}_0 [(V_t - C_t)^+] \quad (4)$$

If we cannot assume independence then we need to condition on default and consider the correlation between the portfolio value and the counterparty default probability. We know that the probability of default in the interval $[t, t + \Delta]$ is given by:

$$P(t < \tau < t + \Delta) = Q_{t+\Delta} - Q_t = Q_t p_t \quad (5)$$

where Q_t is the survival probability process and p_t is the probability of default in $[t, t + \Delta]$ conditional on survival up to t . For small Δ we have $p_t = \lambda_t \Delta$ where λ_t is the counterparty default intensity.

Similarly the expectation of the default probability can be written as:

$$\mathbb{E}_0 [P(t < \tau < t + \Delta)] = \mathbb{E}_0 [Q_t p_t] \quad (6)$$

Putting this together we can write:

$$\mathbb{E}\mathbb{E}(t) = \frac{1}{\mathbb{E}_0 [Q_t p_t]} \mathbb{E}_0 [(V_t - C_t)^+ Q_t p_t] \quad (7)$$

Typically we distinguish between two types of collateral: the variation margin (VM) which we denote by C_t^V and the initial margin (IM) denoted by M_t and in general we have $C_t = C_t^V + M_t$. In the following we will assume that our counterparty is perfectly collateralized with a daily CSA and zero threshold and minimum transfer amount. In this case we can make the typical assumption that the variation margin is given by the value of the portfolio at the last collateral reset date, i.e. $C_t^V = V_t$ and hence:

$$\underline{V_{t+\Delta} - C_t^V = V_{t+\Delta} - V_t \equiv \Delta V_t} \quad (8)$$

With this assumption we have the final expression:

$$\mathbb{E}\mathbb{E}(t) = \frac{1}{\mathbb{E}_0 [Q_t p_t]} \mathbb{E}_0 [(\Delta V_t - M_t)^+ Q_t p_t] \quad (9)$$

3 Leveraged WWR Model

To define our model we need to specify the (scenario dependent) conditional default probability p_t . In typical wrong-way risk models both the default intensity and the portfolio value are modeled as a function of an underlying market process m_t , i.e. $V_t = V(m_t)$ and $\lambda_t = \lambda(m_t)$. As discussed in the introduction we are interested in a different type of relatedness where we have the direct relationship $p_t = p(\Delta V_t)$.

$$m_t \quad V_t = V(m_t) \quad \lambda_t = \lambda(m_t)$$

In particular, we postulate that the counterparty will default with certainty if ΔV_t exceeds a set threshold θ :

$$p_t \equiv p(\Delta V_t, \theta) = \begin{cases} 1 & \Delta V_t \geq |V_t|\theta \\ a & \Delta V_t < |V_t|\theta \end{cases} \quad (10)$$

where θ represents a constant threshold on the relative change in portfolio value. In the case that the change in value falls below the threshold, the default probability is given by a which is a parameter to be calibrated as described below. Note that we require $0 \leq a \leq 1$ and also that typically a will be small, i.e. $a \ll 1$. In our simple model a will also be constant in time.

This choice of default probability is motivated by considering a counterparty with constant leverage. In other words we imagine that the counterparty balance sheet is rebalanced dynamically so that the ratio of assets and liabilities remains constant. This implies that the same relative drop in asset values will lead to insolvency, independent of the total value of the assets. In practice it is the collateral call itself that is likely to be the proximate cause of default.

Portfolio $V \sim \text{Dist}(X)$

We assume that the portfolio value returns follow a distribution X which is independent of V_t and we take to be constant in time for simplicity. The cumulative distribution function for X will be denoted by $F_X(\cdot)$. More specifically, this means that:

$$\Delta V_t \sim |V_t|X \quad F_X(\cdot) \quad (11)$$

and we can write:

$$p(\Delta V_t, \theta) = p(X, \theta) = \mathbb{I}_{X \geq \theta} + a\mathbb{I}_{X < \theta} \quad (12)$$

where we have introduced the standard indicator function $\mathbb{I}$.

We note that p_t is independent of V_t as well as the survival probability Q_t . Hence we have $\mathbb{E}_0[Q_t p_t] = \bar{Q}(t)\bar{p}(t)$, where we have introduced the expected survival probability $\bar{Q}(t) \equiv \mathbb{E}_0[Q_t]$ and expected default probability $\bar{p}(t) \equiv \mathbb{E}_0[p_t]$.

We are thus able to factorise the exposure as:

$$\text{EE}(t) = \frac{1}{\bar{Q}(t)\bar{p}(t)} \mathbb{E}_0[(\Delta V_t - M_t)^+ Q_t p_t] \quad (13)$$

$$= \frac{1}{\bar{Q}(t)\bar{p}(t)} \mathbb{E}_0[Q_t \mathbb{E}_t[(|V_t|X - M_t)^+ p(X, \theta)]] \quad (14)$$

Often, the margin M_t is defined as a percentile in the portfolio value distribution over the margin period of risk. More specifically we can define the margin M_t^q at quantile q and time t by requiring:

$$P(\Delta V_t < M_t^q) = P(|V_t|X < M_t^q) = q \quad (15)$$

and thus:

$$M_t^q = |V_t|m^q \quad (16)$$

with constant $m^q = F_X^{-1}(q)$. If the initial margin is of this type¹ (or zero) then the expected exposure can be written as:

$$\text{EE}(t) = \frac{1}{\bar{Q}(t)\bar{p}(t)} \mathbb{E}_0[Q_t |V_t|] \mathbb{E}[(X - m^q)^+ p(X, \theta)] \quad (17)$$

In general, Q_t will depend on the dynamics of V_s up to the time t . In order to avoid the need for a specific calculation of Q_t we make the simplifying approximation that V_t and Q_t

¹Even in cases where the initial margin has a different definition, it is often conservative to approximate the IM by a quantile in the margin distribution. This is e.g. the case for the the Standard Industry Margin Model (SIMM) which is required by regulation to be larger than a 99% 10 day quantile.

are independent which is sufficient for our purpose. We thus have the final result:

$$\mathbb{E}E(t) = \frac{1}{\bar{p}(t)} \mathbb{E}_0 [|V_t|] \mathbb{E} [(X - m^q)^+ p(X, \theta)] \quad (18)$$

The above expression is convenient as the time dependence is completely captured in the expectation $\mathbb{E}_0 [|V_t|]$ which is in fact closely related to the uncollateralized exposure of the portfolio. The model dependent part is captured in:

$$\mathbb{E} [(X - m^q)^+ p(X, \theta)] = \mathbb{E} [(X - m^q)^+ \mathbb{I}_{X \geq \theta}] + a \mathbb{E} [X^+ \mathbb{I}_{X < \theta}] \quad (19)$$

which is just an expected shortfall calculation when $a = 0$ and $m^q = 0$.

3.1 Calibration

We would like to calibrate the model to the expected default probability of the counterparty. This could be determined by market CDS spreads if we are considering an XVA application or estimated empirically otherwise. We also discuss the case where we do not have a good estimate of the counterparty credit quality.

We hence assume that we are given the forward default intensity curve $\bar{\lambda}(t)$ and hence $\bar{p}(t) = \bar{\lambda}(t)\Delta$ as well as $\bar{Q}(t)$. For simplicity we will assume that the intensity is constant and hence $\bar{p}(t) = \bar{p}$.

The calibration requirement is:

$$\mathbb{E}_0 [Q_t p_t] = \bar{Q}(t) \bar{p} \quad (20)$$

Given the independence of Q_t and p_t in our model this reduces to:

$$\bar{p} = \mathbb{E}_0 [p_t] = \mathbb{E}_0 [\mathbb{I}_{X \geq \theta} + a \mathbb{I}_{X < \theta}] = 1 - (1 - a)F_X(\theta) \quad (21)$$

This can be solved as:

$$a = 1 - \frac{1 - \bar{p}}{F_X(\theta)} \quad (22)$$

Since a is a probability it cannot be negative. This imposes a lower bound on the choice of threshold θ . In particular we can only calibrate to $\bar{p}$ if:

$$\theta \geq \theta^* \equiv F_X^{-1}(1 - \bar{p}) \quad (23)$$

The intuition is that if the threshold is too low then the the probability of exceeding it is always larger than the default probability even if setting $a = 0$. Setting $\theta = \theta^*$ and hence $a = 0$ will maximize the WWR exposure for a given default probability $\bar{p}$. If on the other hand we set $\theta = \infty$ then $a = \bar{p}$ and there will be no WWR in the model.

Alternatively, if we do not have a reliable external estimate of $\bar{p}$ but we do have a strong view on the threshold θ then we can estimate $\bar{p}$ within the model by setting $a = 0$. We are thus assuming that the counterparty will default if and only if the change in portfolio value is greater than the threshold. In this case we have simply:

$$\bar{p} = P(X \geq \theta) = 1 - F_X(\theta) \quad (24)$$

and

$$\mathbb{E}E(t) = \mathbb{E}_0 [|V_t|] \frac{\mathbb{E} [(X - m^q)^+ \mathbb{I}_{X \geq \theta}]}{\mathbb{E} [\mathbb{I}_{X \geq \theta}]} = \mathbb{E}_0 [|V_t|] \mathbb{E} [(X - m^q)^+ | X \geq \theta] \quad (25)$$

which is entirely determined by the portfolio dynamics and the choice of threshold.

3.2 Exposure Ratios

In order to assess the magnitude of the expected exposure in the presence of WWR we can take ratios of different exposure quantities. This also allows us to remove the term dependency in the expectation. We now provide the expressions for three useful cases.

Unconditional Expected Exposure: The expected exposure in the absence of wrong way risk is a special case of our expression in (18) in the limit that $\theta \rightarrow \infty$ and hence $a = \bar{p}$. We have:

$$\text{EE}^U(t) = \mathbb{E}_0 [|V_t|] \mathbb{E} [(X - m^q)^+] \quad (26)$$

Uncollateralized Expected Exposure: In the uncollateralized case the expected exposure is simply:

$$\text{EE}^{UC}(t) = \mathbb{E}_0 [V_t^+] = \frac{1}{2} \mathbb{E}_0 [|V_t|] \quad (27)$$

Expected Margin: Perhaps the most useful reference point is the expected margin $\bar{M}^q(t)$ at quantile q . This is defined as the expectation:

$$\bar{M}^q(t) \equiv \mathbb{E}_0 [M_t^q] = \mathbb{E}_0 [|V_t|] m^q \quad (28)$$

The ratio of $\text{EE}(t)$ to any of the above quantities will not depend on V_t and thus be constant in time. This allows us to focus on the calculation of $\text{EE}(t)$ for a single time point. It also means that we can implement adjustments for WWR by simple scaling. In particular, we know that CVA is linear in the expected exposure. We can hence write CVA in the presence of wrong way risk as:

$$\text{CVA} = \alpha \text{CVA}^U \quad (29)$$

where CVA^U is the CVA ignoring WWR and α is a constant given by:

$$\alpha = \frac{\text{EE}(t)}{\text{EE}^U(t)} = \frac{\mathbb{E} [X^+ p(X, \theta)]}{\bar{p} \mathbb{E} [X^+]} \quad (30)$$

assuming for simplicity that there is no initial margin.

We show in appendix A that in the case that the distribution of X is a student's t distribution and $a = 0$ then the exposure ratios are also independent of the volatility of X and depend only on the tail behaviour of the distribution.

As an aside, our set up also allows for the simple estimation of the expected exposure subject to dynamic initial margin as a constant scalar times the unmargined exposure, i.e.:

$$\mathbb{E}_0 [(\Delta V_t - M_t^q)^+] = \beta \mathbb{E}_0 [(\Delta V_t)^+] \quad (31)$$

where $\beta = \mathbb{E} [(X - m^q)^+] / \mathbb{E} [X^+]$ is a simple function of the distribution of X . This is just a special case of the results presented in Anderson et. al. in [6].

4 Results

In this section we will apply the model introduced above. We first investigate the sensitivity of the model to the key parameters to gain intuition around the model behaviour. Subsequently, we will study a real-world test case in order to assess whether the model output is realistic.

4.1 The Portfolio Distribution

To evaluate our model we need to make a choice for the distribution of X . It is well-known that financial distributions are heavy tailed and a common choice for modelling risk factor returns is the student's t-distribution.

The student's t-distribution is denoted T_ν with density $t_\nu(\cdot)$ and is parameterized by the degrees of freedom ν , which governs the tails of the distribution. When $\nu \rightarrow \infty$ then $t_\nu(\cdot)$ converges to the normal distribution. In practice values of $\nu < 10$ indicate heavy tails and we require $\nu > 2$ for the variance of our distribution to be defined. The estimation of ν for financial time series can be performed empirically and has been well-studied in the literature.

For this article we have analyzed a five year history (August 2017 to August 2022) of the weekly log-returns of four main risk factors: rates (USD IR), FX (GBP/USD), equities (S&P 500) and commodities (Natural Gas). The best fit student's t distribution as well as the estimation ν for each risk factor is shown in figure 1. We observe that ν varies between 2.6 and 3.3 which indicates significantly heavy tails and is consistent with similar studies in the literature (c.f. [2]).

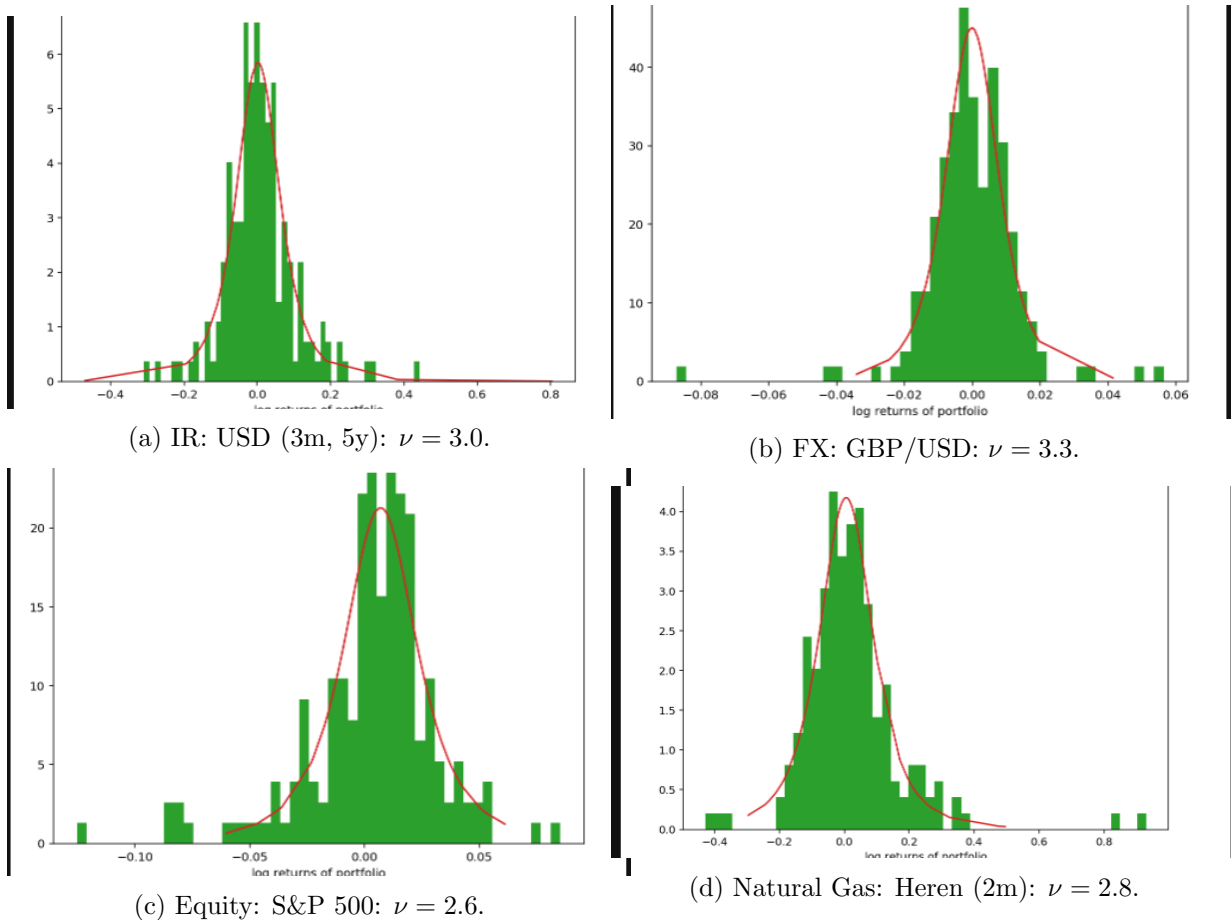


Figure 1: Estimation of ν in different markets.

Parameter	Default Value
Default Probability: $\bar{p}$	300 bps
Margin Period of Risk: Δ	5 days
Volatility over MPoR (annualized): σ	30%
Degrees of freedom: ν	3
Margin quantile: q	99%
Threshold: θ	$\theta = \theta^*$

Table 1: Default values for the main parameters.

Base Exposure	IM?	EE WWR / Base
Unconditional EE (no WWR)	N	33.0
Expected Margin (99%)	N	4.0
Uncollateralized EE	N	0.9
Unconditional EE (no WWR)	Y	24.8
Expected Margin (99%)	Y	3.0
Uncollateralized EE	Y	0.7

Table 2: Ratios of WWR Expected Exposure over benchmark exposure metrics.

4.2 Parameter Sensitivity

We now examine the ratio of the WWR exposure to the benchmark exposure metrics described in section 3.2. For this purpose we set $X \sim \hat{\sigma}\sqrt{\Delta}T_\nu$ in which case the mean of X is zero and the scale parameter $\hat{\sigma}$ is related to the (annualized) volatility σ of X by:

$$\sigma = \hat{\sigma}\sqrt{\frac{\nu}{\nu-2}} \quad (32)$$

Our default settings are listed in table 1 and the corresponding exposure ratios in table 2. We consider the expected exposure both in the presence of initial margin (at a 99% quantile) and without. As expected, we observe that WWR has a significant impact. In the unmarginated case, the ratio of the WWR exposure to the unconditional exposure (no WWR) is at 33 and the ratio to the expected margin is 4. In fact, the WWR exposure is close to the uncollateralized expected exposure. In the margined case, the ratios are slightly lower but still significant.

In the following we discuss the sensitivity of the ratio of the WWR exposure over the expected margin to the main model parameters. We assume that exposure is margined at 99% quantile.

Threshold In figure 2a we plot the exposure ratio as a function of the threshold in units of the volatility over the margin period of risk. We do this for different levels of the counterparty default probability. As expected the ratio converges to 1 as the threshold is increased. However, even if the threshold is 30 times the MPoR volatility the exposure still exceeds the initial margin by several multiples in particular for counterparties with low default probabilities. We also observe that there is a minimum threshold in our model which decreases as the default probability is increased. This is because a lower threshold

implies a higher probability of exceeding the threshold and hence a higher attainable default probability. We note that the minimum threshold in the case of $\nu = 3$ and $\bar{p} = 300\text{bps}$ is around eight times the MPoR volatility. In other words, a probability of default of 300bps corresponds to market moves that are eight times the standard deviation.

Default Probability We plot the exposure ratio as a function of the default probability for various levels of ν in figure 2b. As expected, the ratio reduces as $\bar{p}$ increases which is a standard feature in WWR models. For a highly rated counterparty to default we require more extreme market moves. This means that *conditional* on default the exposure will be high.

Margin Period of Risk The exposure decreases as the margin of period of risk is increased as is illustrated in figure 2c. This is because the the probability of default in the margin period $\bar{p}$ increases linearly with Δ . The MPoR volatility σ however only increases by the square root of Δ . This means that $\bar{p}$ is effectively increased for longer margin periods of risk. Since a larger expected default probability implies as smaller WWR effect this leads to the observed result.

Degrees of Freedom We show in appendix A that in the case that the threshold is set to the minimum (i.e. $a = 0$), the exposure ratio is independent of the volatility of the portfolio distribution over the margin period of risk. This means that the ratio is entirely driven by the tail behaviour determined by the degrees of freedom ν . In figure 2d we see that the exposure ratio decreases as ν increases and the distribution converges to a Gaussian. However, even for high levels of ν the WWR exposure can still exceed the posted margin. As indicated in section 4.1 markets exhibit heavy-tailed behaviour and empirical estimates of ν are around three. Considering jump processes would likely increase the WWR effect even further which means that the ratios we provide for low values of ν represent conservative estimates of WWR.

4.3 Case Study

We now consider a practical application of our model to a portfolio similar in characteristics to that of the Archegos Capital Management hedge fund which defaulted in March 2021. Archegos was a highly leveraged fund focusing on long-short equity strategies. Its default was caused by a large and sudden drop in the value of the fund's main underlying equity positions and the associated collateral calls. The eventual total default loss across all of Archegos' brokers is estimated at \$10 billion with one of the largest single portfolio losses being that of Credit Suisse at \$5.5 billion as detailed in the report [1].

We thus consider an equity portfolio with key parameters given in table 3. The portfolio has a net value of \$7 billion. The annualized volatility is set at 70% and we assume that the counterparty has a default probability of 300bps which corresponds roughly to a B+ rating. The portfolio is also subject to variation margin as well as initial margin which we assume to be fixed at \$500 million prior to default. These levels are chosen to be comparable those reported for Archegos in [1].

We now look at credit exposure calculations for this portfolio both using a typical model ignoring WWR and using our model which assumes that the default of the counterparty is linked to the size of potential market moves. We will see that the incorporation of WWR

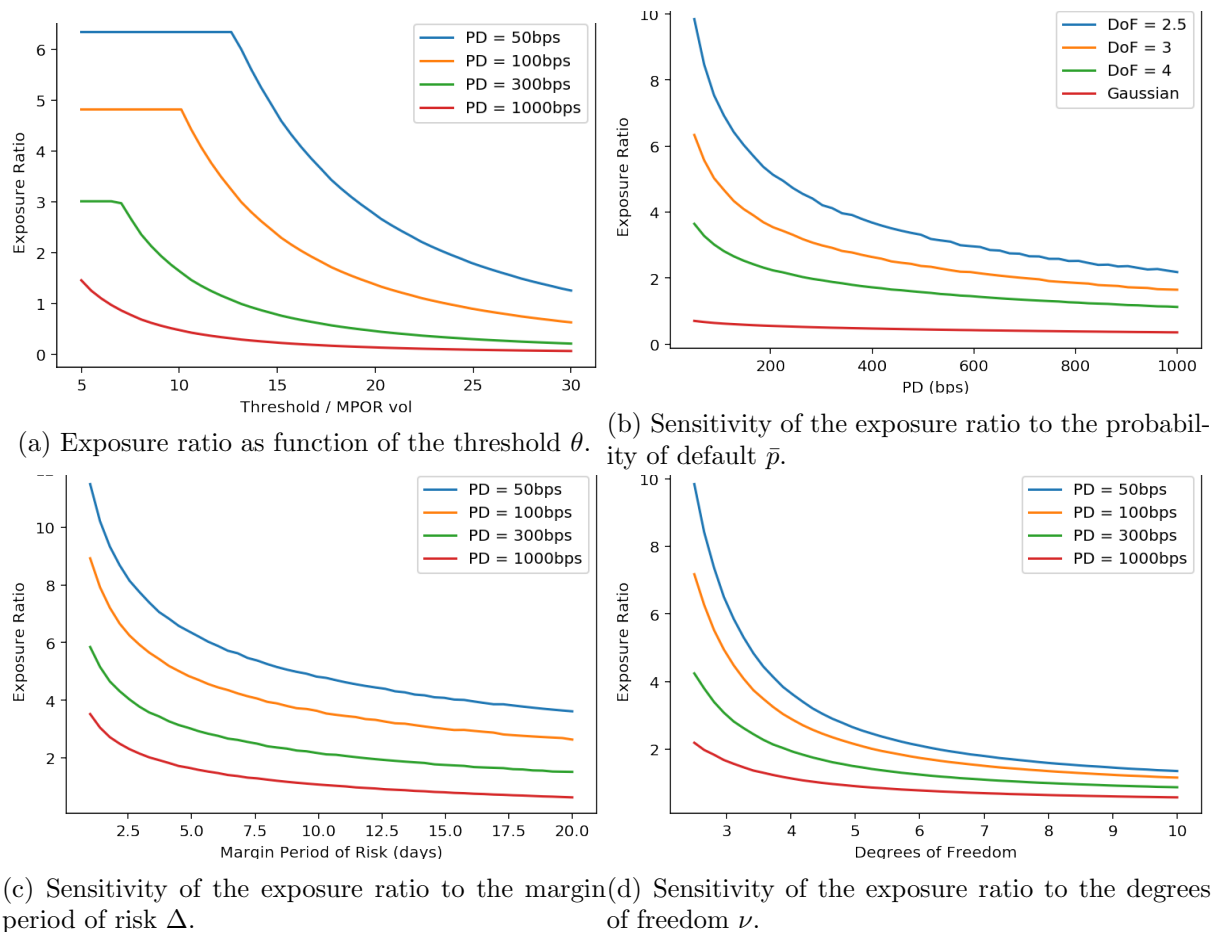


Figure 2: Sensitivity of the exposure ratio to the model parameters.

is key for obtaining realistic levels of exposure which should be compared to the \$5.5b loss reported on Credit Suisse Achegos portfolio .

We make the simplifying assumption that we can treat our entire portfolio as a single stock position with value equal to the net position of \$7 billion. This means effectively that we are assuming perfect correlation between all equities. This will provide a lower bound for our exposure estimates², which as we shall see is already significant. It is straightforward to adapt this test case to any particular portfolio of interest.

*= \$7b
single stock*

Hence we model our portfolio with a log-student's t-distribution i.e:

$$X \sim e^{\hat{\sigma}\sqrt{\Delta}T_{\nu} - \frac{1}{2}\hat{\sigma}^2\Delta} \quad (33)$$

Since we are interested in absolute exposure quantities and not ratios we also need to specify the distribution of the portfolio value V_t . However, here we are focused on the short term exposures and $t \ll 1$. This means we can approximate the term volatility of V_t to zero and set $V_t = V_0$. The starting point for our portfolio value process is hence at the net value of $V_0 = \$7b$. We assume that the margin period of risk over which the portfolio is closed out post default is 5 business days.

²This is because a more disperse portfolio can have losses greater than the net value of \$7 billion. Note that we are fixing the portfolio volatility at 70%.

Portfolio metric	Value
Net market value	\$7b
Default probability	300bps (B+)
Volatility (annualized)	70%
Initial Margin	\$500
MPoR	5 days

Table 3: Key portfolio characteristics.

We are now ready to calculate the expected exposure and PFE³ with WWR and without. The results of our model assuming an unconditional default probability of $\bar{p} = 300\text{bps}$ are given in table 4a. We see that the expected exposure at \$56 million is negligible in the case that WWR is ignored. The expected loss which can be viewed as a proxy for the CVA is calculated here as the product of the default probability and the expected exposure. This is also small in the unconditional case (\$1.7 million). The unconditional PFE is equal to \$411 which is also low. When introducing WWR via our model we get realistic levels of the expected exposure and PFE (\$3.4 and \$5.5 billion respectively) which can be compared to observed Archegos loss of \$5.5 billion.

We also calculate the minimum threshold to be \$3.5 billion which is high compared to what we expect a counterparty to be able to fund in terms of collateral. We note that the Archegos default coincided with a drop in the Archegos portfolio value of around \$2.5b. The high level of threshold is a consequence of assuming a default probability of 300bps combined with the high volatility of the portfolio. Given that the default probability is typically only a crude estimate, this is an example where fixing a threshold and estimating the default probability endogenously might be more suitable.

In figure 3 we plot the threshold against the probability of default. We note that to a first approximation knowledge of the threshold already tells us the expected exposure which is just an expected shortfall calculation and thus slightly above the threshold.

If we want the minimum threshold to be around \$2.5b then the expected default probability needs to be at ca. $\bar{p} = 12\%$. In this case we obtain a lower expected exposure of \$2.4 billion and a PFE of \$4.4 billion as shown in table 4b. The expected loss is now higher at \$320 million given the elevated default probability. These results still represent reasonable exposure estimates in line with experience. We note that a comprehensive assessment of the expected loss or PFE would also need to also take into consideration how large the portfolio is compared to the market. Adjusting estimates for lack of liquidity would increase exposures further.

5 Conclusion

We have presented a simple model of wrong-way risk (WWR) exposure for leveraged and collateralized counterparties. This is applicable whenever the counterparty solvency is directly linked to the performance of its portfolio which is typical for hedge funds.

Leveraged counterparties are vulnerable to large market moves and subsequent collateral

³The PFE conditional on default can be calculated simply in the approximation that $V_t = V_0$ as shown in appendix B. It is given by the unconditional PFE at a modified quantile $q' \gg q$.

Metric	Uncond	WWR	Ratio	Metric	Uncond	WWR	Ratio
PFE	411	5456	13	PFE	411	4389	11
EE	56	3795	68	EE	56	2664	47
EL (PD · EE)	1.7	114	68	EL (PD · EE)	6.7	320	47
Threshold	NA	3509	NA	Threshold	NA	2436	NA

(a) PD = 300bps.

(b) PD = 1200bps.

Table 4: Modelled exposure metrics for the case study portfolio (\$m).

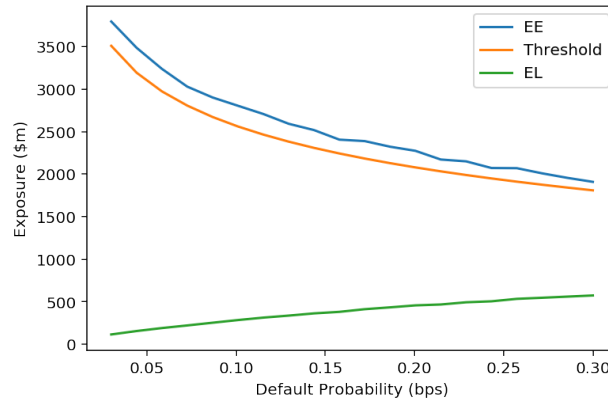


Figure 3: Exposure metrics as a function of the default probability.

calls and hence we model the counterparty default probability as a function of the change in portfolio value over the margin period of risk. The key input parameter is the threshold on the maximum collateral that a counterparty will be able to fund.

Adjustments to the expected exposure are straightforward to calculate and implement in practice. We show that the incorporation of WWR has a large impact on exposures in line with previous results in the literature. For realistic parameter settings, our model results in expected exposures that are around 30 times the unconditional exposure and four times the expected initial margin calculated at the 99th percentile.

We consider the recent Archegos default as a test case for our model. Archegos was a highly leveraged equity hedge fund which sustained significant losses due to a sudden drop in the equity market. This caused the eventual default of the fund in March 2021. Given the highly collateralized nature of the portfolio, standard exposure models predicted a minimal expected loss as well as potential future loss prior to default.

Our model allows us to recover realistic estimates of the exposure which are of similar magnitude to the realized losses. This requires only minimal model input as the exposure profiles are largely driven by the tails of the portfolio return distribution. For reasonable distributional estimates and default thresholds we find that WWR increases the expected exposure by around 50 times compared to a model without WWR and is thus key to understanding the credit risk posed by leveraged and collateralized counterparties.

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A Independence of Exposure Ratios on Volatility

Here we demonstrate that the exposure ratios are independent of the volatility of X when the threshold is set to the minimum level (i.e. $a = 0$) and X follows a student's t distribution as defined in section 4.1.

We consider the expectation:

$$\mathbb{E} [X^+ p(X, \theta^*)] = \mathbb{E} [X^+ \mathbb{I}_{X \geq \theta}] = \int_{\theta^*}^{\infty} x^+ f(x) dx \quad (34)$$

where θ^* is the minimum threshold calculated for scale $\hat{\sigma}$, i.e.:

$$\theta^* = F_X^{-1}(1 - \bar{p}) = \hat{\sigma} T_{\nu}^{-1}(1 - \bar{p}) \quad (35)$$

where $T_{\nu}^{-1}(\cdot)$ is the inverse cumulative student's t distribution. Substituting in the explicit form of the distribution we have:

$$\mathbb{E} [X^+ p(X, \theta^*)] = \int_{\frac{\theta^*}{\hat{\sigma}}}^{\infty} (\hat{\sigma} z)^+ t_{\nu}(z) dz \quad (36)$$

$$= \hat{\sigma} \int_{T_{\nu}^{-1}(1 - \bar{p})}^{\infty} z^+ t_{\nu}(z) dz \quad (37)$$

Both the unconditional expected exposure and the expected margin are also proportional to the scale $\hat{\sigma}$. This means that the ratios of $EE(t)$ to $EE^U(t)$ or $M^q(t)$ are independent of the volatility of X and depend only on the degrees of freedom ν .

B Wrong-way Risk PFE

In the presence of wrong-way risk, the PFE at time t , $\mathcal{P}_t(q)$, is defined as the percentile q in the exposure distribution $E_t = (V_t - C_t)^+$ conditional on default, i.e.:

$$P(E_t > \mathcal{P}_t(q) | \tau = t) = 1 - q \quad (38)$$

We now show that when $a = 0$ and $V_t = V_0$, then $\mathcal{P}_t(q)$ can be calculated as an unconditional PFE, $\mathcal{P}_t^U(q')$, at a higher quantile q' .

Using Bayes rule we have:

$$1 - q = P(\tau = t | E_t > \mathcal{P}_t(q)) \frac{P(E_t > \mathcal{P}_t(q))}{P(\tau = t)} \quad (39)$$

The distribution of E_t is given by $V_0 X$. We know that the PFE conditional on default will be larger than the default threshold θ . Hence, given that $a = 0$, we know that $P(\tau = t | E_t > \mathcal{P}_t(q)) = 1$. This implies:

$$P(E_t > \mathcal{P}_t(q)) = (1 - q)P(\tau = t) = \mathcal{P}^U(q') \quad (40)$$

with

$$1 - q' = (1 - q)P(\tau = t) \approx \bar{p} \quad (41)$$